

Analysis of LISA Noise Curves and Characteristic Strains for Gravitational Wave Sources

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Abstract

This project investigates the sensitivity limitations of the Laser Interferometer Space Antenna (LISA), a planned space-based gravitational wave observatory. By modeling the theoretical noise power spectral densities, we compare with the observational window for LISA compared to ground-based counterparts like LIGO. The study further analyzes the characteristic strains of various astrophysical sources—including Compact Binary Coalescences (CBCs) and stochastic backgrounds—to determine their detectability within LISA’s operating frequency band (10^{-4} Hz to 1 Hz).

Contents

1	Introduction	3
1.1	Gravitational Wave Sources	3
1.2	Detector Frequency Regimes	3
2	Methodology	4
2.1	LIGO Sensitivity Model	4
2.2	LISA Sensitivity Model	4
3	Results and Analysis	5
3.1	Simulated Noise Curves	5
3.2	Source Detectability	5
4	Conclusion	5

1 Introduction

Gravitational waves (GWs) have opened a new window onto the universe, allowing us to observe cosmic phenomena through the ripples in spacetime they produce. Predicted by Einstein in 1916, GWs were first directly detected in September 2015 by the Advanced LIGO detectors. This landmark detection, arising from a binary black hole merger, confirmed a century-old prediction and marked the birth of gravitational wave astronomy. Since then, dozens of compact object mergers (black holes and neutron stars) have been observed via their GW “chirp” signals – waveforms that increase in frequency and amplitude as the two objects [spiral together](#). These observations not only validate general relativity but also provide unprecedented insights into the properties of black holes and neutron stars.

While the current observations have been limited by the sensitivity of LIGO-VIRGO-KAGRA (LVK) collaboration, which measure GW in the frequency range 20 Hz to 20,000 Hz with sensitivity reaching to 10 Hz in the upcoming years, the future of GW astronomy will include space-based GW detectors like LISA which will be sensitive in the mhz band. These detectors will open up window to an entire new ways to study compact objects and specifically types of binaries which are not visible in the sensitivity range of LIGO - Binary white dwarfs binaries (**BWDs**),

1.1 Gravitational Wave Sources

Gravitational wave sources are generally categorized by their signal morphology:

- **Compact Binary Coalescences (CBCs):** These signals arise from the inspiral, merger, and ringdown of compact objects. They include Binary Neutron Stars (BNS), which may have electromagnetic counterparts (e.g., GW170817), and Binary Black Holes (BBH), such as the historic GW150914 event.
- **Continuous Waves:** Persistent, monochromatic signals generated by rotating neutron stars (pulsars) possessing a quadrupole moment due to structural ellipticity. These are prime targets for long-integration searches.
- **Stochastic Background:** A random GW field produced by the superposition of unresolved astrophysical sources or primordial processes from the early universe.
- **Burst Sources:** Unmodelled, transient bursts of energy from cataclysmic events like supernovae or cosmic string cusps.

1.2 Detector Frequency Regimes

The detectability of a source depends on the detector’s sensitive bandwidth:

1. **Ground-Based (LIGO/Virgo):** Sensitive between 10 Hz and 1000 Hz. Optimized for stellar-mass binary mergers.
2. **Space-Based (LISA):** Sensitive between 10^{-4} Hz and 1 Hz. Optimized for Super-massive Black Hole (SMBH) binaries and Extreme Mass Ratio Inspirals (EMRIs).
3. **Pulsar Timing Arrays (NANOGrav):** Sensitive to nanohertz frequencies, detecting the background hum of merging galaxies.

4. **DECIGO and decihz detectors:** Sensitive to inspiraling NS-WD frequencies in the decihz range - 0.1 to 1 Hz

2 Methodology

The sensitivity of a detector is characterized by its Noise Power Spectral Density (PSD), $S_n(f)$. For a signal to be resolved, its characteristic strain, $h_c(f)$, must generally exceed the detector's effective noise strain, $h_n(f) = \sqrt{f S_n(f)}$.

2.1 LIGO Sensitivity Model

The Advanced LIGO noise curve is modeled analytically. The Noise PSD, $S_n(f)$, is approximated by contributions from seismic noise (low-f), thermal noise (mid-f), and shot noise (high-f). The reference PSD is given by:

$$S_n(f) = 10^{-49} \left(\frac{f}{f_0} \right)^{-4.14} + 10^{-50} \left(\frac{f}{f_0} \right)^{-0.69} + 10^{-52} \left(\frac{f}{f_0} \right)^{2.145} \quad (1)$$

where $f_0 = 215$ Hz.

2.2 LISA Sensitivity Model

LISA's noise budget is derived from the ESA mission specifications, prominently featured in Robson et al. (2019). The total noise consists of three primary components:

1. **Optical Metrology Noise (P_{oms}):** Arising from shot noise and laser phase fluctuations.

$$P_{oms}(f) = (1.5 \times 10^{-11})^2 \left[1 + \left(\frac{2 \times 10^{-3}}{f} \right)^4 \right] \quad (2)$$

2. **Acceleration Noise (P_{acc}):** caused by residual forces on the test masses.

$$P_{acc}(f) = (3 \times 10^{-15})^2 \left[1 + \left(\frac{0.4 \times 10^{-3}}{f} \right)^2 \right] \left[1 + \left(\frac{f}{8 \times 10^{-3}} \right)^4 \right] \quad (3)$$

3. **Confusion Noise (S_c):** An unresolved background from galactic white dwarf binaries.

$$S_c(f) = A f^{-7/3} e^{-f^\alpha + \beta f \sin(\kappa f)} [1 + \tanh(\gamma(f_k - f))] \quad (4)$$

The total effective Noise PSD for LISA is aggregated as:

$$S_n(f) = \frac{10}{3L^2} \left(P_{oms}(f) + \frac{4P_{acc}(f)}{(2\pi f)^4} \right) \left(1 + 0.6 \left(\frac{f}{f_*} \right)^2 \right) + S_c(f) \quad (5)$$

where $L = 2.5 \times 10^9$ m is the arm length and f_* is the transfer frequency.

3 Results and Analysis

3.1 Simulated Noise Curves

The simulation generates sensitivity curves for both detectors. The resulting plots (Figures 1 and 2) reveal the distinct "bucket" shape of the sensitivity curves, where the minimum represents the frequency of optimal sensitivity.

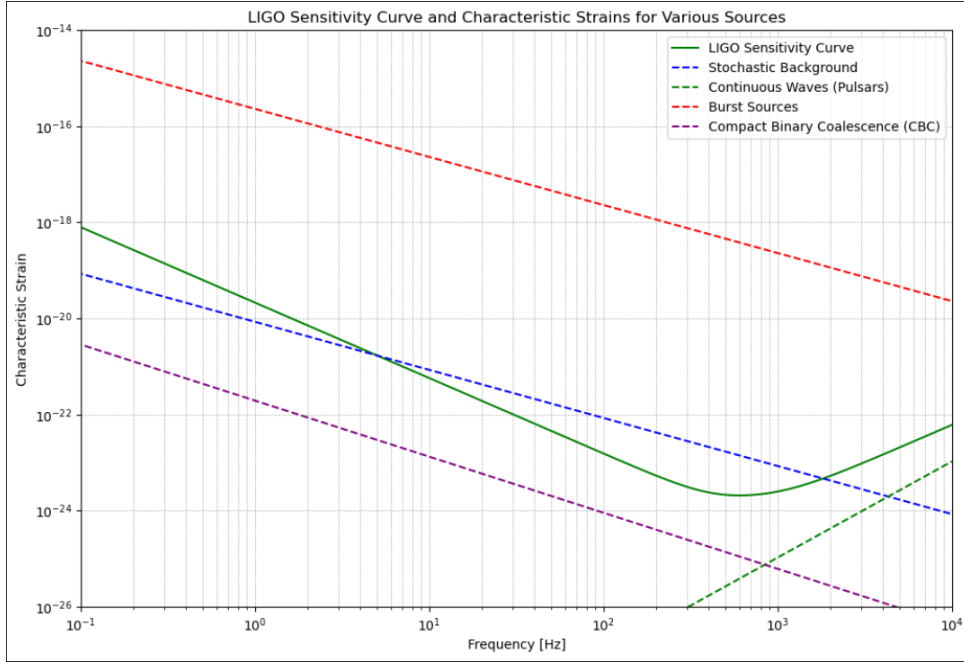


Figure 1: Simulated LIGO sensitivity curve plotted against characteristic strains of various sources. (Figure representation based on report analysis).

3.2 Source Detectability

The analysis compares the characteristic strain $h_c(f)$ of sources against the sensitivity curves $\sqrt{fS_n(f)}$.

- **LIGO Context:** The simulation indicates difficulty in detecting continuous waves and stochastic backgrounds unless the signal amplitude is significantly high. However, Compact Binary Coalescences (CBCs) show strains well above the noise floor in the 100 Hz range.
- **LISA Context:** LISA shows exceptional sensitivity to massive binaries. However, typical galactic pulsars fall below the LISA sensitivity curve, rendering them undetectable. Burst sources appear above the curve, indicating potential detectability depending on distance and energy output.

4 Conclusion

This study highlights the complementary nature of ground and space-based detectors. While LIGO opens the window to stellar-mass mergers, the LISA noise curve analysis

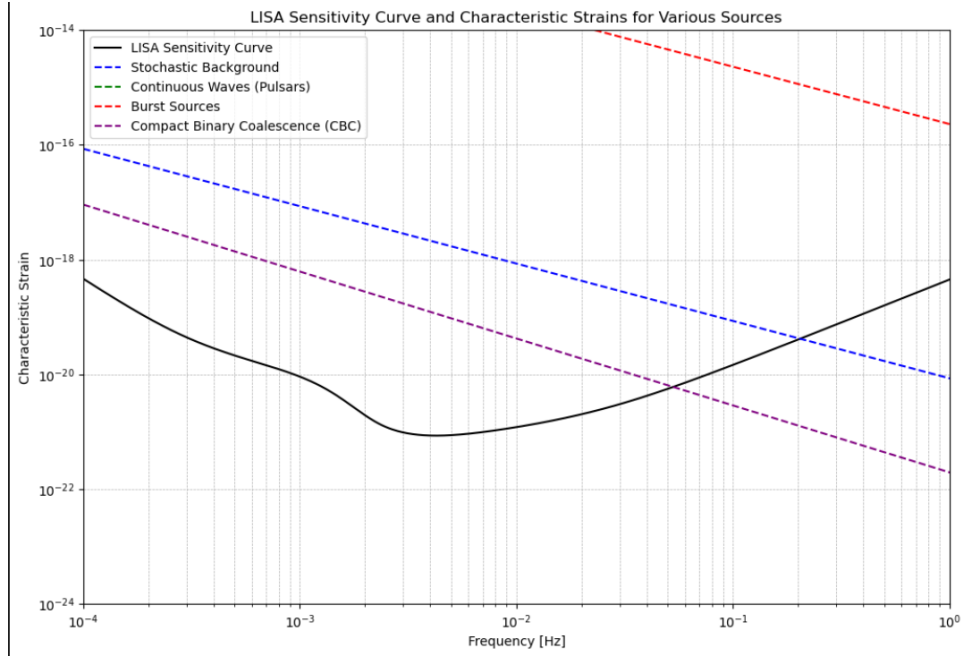


Figure 2: Simulated LISA sensitivity curve. Note the "confusion noise" limit at lower frequencies.

confirms its capability to probe the low-frequency universe, essential for understanding supermassive black hole evolution. Optimizing the noise budget—specifically reducing acceleration noise at low frequencies—remains critical for the mission's success.

References

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